

# PCIS Environmental & Ventilation Report

House: Farm A - House 3 | Generated: 2026-07-20 15:49 UTC

## Flock Summary

| Parameter                  | Value    |
|----------------------------|----------|
| Breed (informational only) | Ross 308 |
| Bird count                 | 20,000   |
| Body weight                | 2.50 kg  |

## Environmental Conditions

| Parameter                        | Value          |
|----------------------------------|----------------|
| Indoor temperature               | 29.0 °C        |
| Indoor relative humidity         | 60 %           |
| Outdoor temperature              | 38.0 °C        |
| Outdoor relative humidity        | 30 %           |
| Supply air (post-pad if pads ON) | 27.2 °C / 74 % |

## Comfort Assessment

| Parameter                          | Value              |
|------------------------------------|--------------------|
| Target temperature (Aviagen table) | 20.7 °C            |
| Deviation from target              | +8.3 °C            |
| Temperature-humidity index (THI)   | 28.1 (heat_stress) |
| Bird Comfort Index (0-100)         | 0                  |

Note: the Bird Comfort Index is PCIS's own composite synthesis of the target-temperature deviation and THI metrics above, not a published/validated instrument. See `comfort_engine.py` for the individually-cited sources of its component metrics.

## Recommendation

| Parameter            | Value         |
|----------------------|---------------|
| Fans ON              | 5             |
| Cooling pads ON      | Yes           |
| Required airflow     | 171,006 m³/h  |
| Governing constraint | Sensible Heat |
| Confidence score     | 70 / 100      |

## Engineering Explanation

- Flock of 20000 birds @ 2.5 kg produces 161.8 kW sensible / 260.5 kW latent heat and 374.9 kg/h moisture [CIGR (2002) via bird\_metabolism.py].
- Envelope conduction: -7.29 kW (positive = loss to outside, negative = heat gain from outside).
- Net sensible load ventilation must handle: 169.1 kW.
- Outdoor temp (38.0C) exceeds the comfort target (20.7C) by more than 1.0C -- plain ventilation cannot reach target, so pads are recommended ON. Pad-leaving supply air: 27.2C / 74% RH (assumed 75% saturation efficiency).
- -15 confidence: pad efficiency is a single design-point assumption from MSU Extension guidance, not a manufacturer velocity-dependent curve (see cooling\_pad.py docstring).
- -10 confidence: using the default 420 ppm outdoor CO2 background rather than a locally measured value.
- Airflow requirements (m3/h): sensible\_heat=171,006, co2=27,007, minimum\_ventilation=34,350 -> governing constraint: sensible\_heat (171,006 m3/h).
- Big Dutchman V130-3-1.5 PS (no cone, 1.5 HP motor) delivers 40,700 m3/h at 30 Pa -> 5 fan(s) needed.
- -5 confidence: comfort\_index is PCIS's own composite synthesis, not a validated published instrument (see comfort\_engine.py docstring).
- Comfort assessment: target=20.7C, deviation=+8.3C, THI=28.1 (heat\_stress), comfort\_index=0/100.

Generated by PCIS (Poultry Climate Intelligence System). Figures marked as assumptions or design-point estimates in the explanation above should be verified against site-specific data before being relied on for critical decisions. See PROGRESS.md in the project repository for full source citations and known limitations of each underlying model.